

Particulate Matter

Monitoring & Health Effects

Issue 2026-08-21 | Entry window 21 Aug 2026 | 19 in scope, 46 screened out

19RECORDS IN SCOPE
(46 SCREENED OUT)**7**OF 10 CLUSTERS
OCCUPIED**8**EXPOSURE /
MODELLING (42%)**3**PREPRINTS
THIS ISSUE**1**RISK ESTIMATE
ON THE PLOT

Signal of the day

Three independent papers today answer the same question — *which part of the mass?* — and none of them answers it with the mass.

Wei et al. (*Environ Res*) ran generalized weighted quantile sum regression over **15 PM_{2.5} chemical components** in 528,127 SEER-Medicare patients aged ≥ 65 with lung cancer, and report **RR 1.011 (95% CI 1.010–1.013) per decile increase in the mixture**, with silicon, nitrate, vanadium, zinc and iron carrying the largest weights — a road-dust, traffic and heavy-fuel-oil signature rather than a mass signature. Chen et al. (*Environ Toxicol Pharmacol*) decompose PM_{2.5} into five constituents in 24,940 UK Biobank participants and find *ammonium* the strongest single predictor of incident depression, with 140 plasma proteins and 3 metabolites mediating up to 8.0%. And Chai et al. (*Environ Pollut*) show why that matters upstream: in suburban Nanjing, secondary inorganic aerosol is **49.55% of PM_{2.5} mass**, nitrate-dominated and ammonia-buffered, so *PM_{2.5} mass barely responds to moderate NH₃ cuts* — it takes $>60\%$ reduction to move, whereas cutting HNO₃ alone performs as well as cutting both. The common implication for instrumentation is direct: a monitor that reports only mass cannot distinguish the exposure that carries the risk from the exposure that does not, and cannot tell a policy-maker which precursor to regulate.

What changed today

- ▶ **The issue is unusually weighted toward exposure and deposition.** Eight of nineteen records (42%) sit in *Exposure assessment & modelling*, and four of those eight measure PM by what it *lands on* rather than what it does in air — pine needles in Milazzo, privet leaves after two Albanian fires, ten tree species in Delhi, and Elsevier's aerosol–precipitation pair. Deposition biomonitoring is cheap, retrospective and spatially dense; it is also unstandardised, and no two of these four papers report leaf-loading in the same units.
- ▶ **An airborne-microplastic baseline for Southeast Asia.** Kan et al. report the first Phnom Penh measurements: **$0.97 \pm 0.30 \text{ MP m}^{-3}$ in TSP and $0.33 \pm 0.10 \text{ MP m}^{-3}$ in the PM_{2.5} fraction**, 82% below 30 μm Feret diameter. Roughly a third of the airborne microplastic count sits inside the size fraction that conventional PM_{2.5} monitoring already samples — and that no conventional PM_{2.5} monitor identifies.
- ▶ **An ammonia-rich regime where the obvious control lever does not work.** The Nanjing thermodynamic sensitivity analysis is total-nitrate-limited, not ammonia-limited. This is a concrete counter-example to the assumption that NH₃ abatement is a general-purpose PM_{2.5} control.
- ▶ **Indoor infiltration quantified at a useful sample size.** A preprint covering **48 residences across eight northern Chinese cities**, four seasons, 32,256 hourly paired indoor/outdoor observations, gives an infiltration factor of **0.28 ± 0.15** and attributes **57–69% of indoor PM_{2.5} to outdoor sources**, peaking in the heating season. Air-purifier operation was the single strongest determinant in the model ($\beta = -0.935$).
- ▶ **Mechanism: a specific innate-immune axis, with a pharmacological rescue.** Yan et al. close the loop from 21-day PM_{2.5} exposure through mitochondrial damage and mtDNA release to cGAS–STING activation and macrophage M1 polarisation, and show that the cGAS inhibitor RU.521 attenuates the whole chain in vivo.
- ▶ **One estimate, not zero — but it is small and it is a mixture.** After yesterday's zero-effect issue, the forest plot is populated again, with a single row. See the note on the figure: the axis is now floored at a $1.6\times$ span precisely so a 1.1% risk ratio is not drawn as though it were a large one.

Corpus analytics

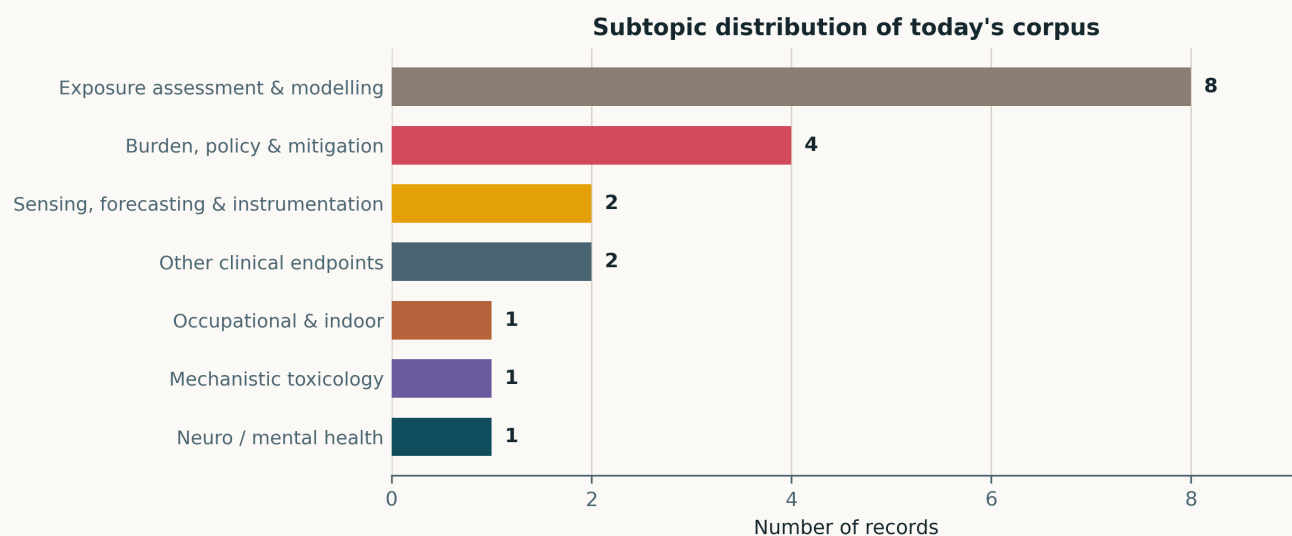


Fig. Subtopic assignment of the 19 in-scope records; single-label by dominant contribution. *Exposure assessment & modelling* takes 8 of 19, more than twice any other cluster, and *Burden, policy & mitigation* takes 4. Three of the ten canonical clusters are empty today — cardiovascular/metabolic, respiratory/allergic and reproductive/developmental — which is the widest health-side gap since 8 August.

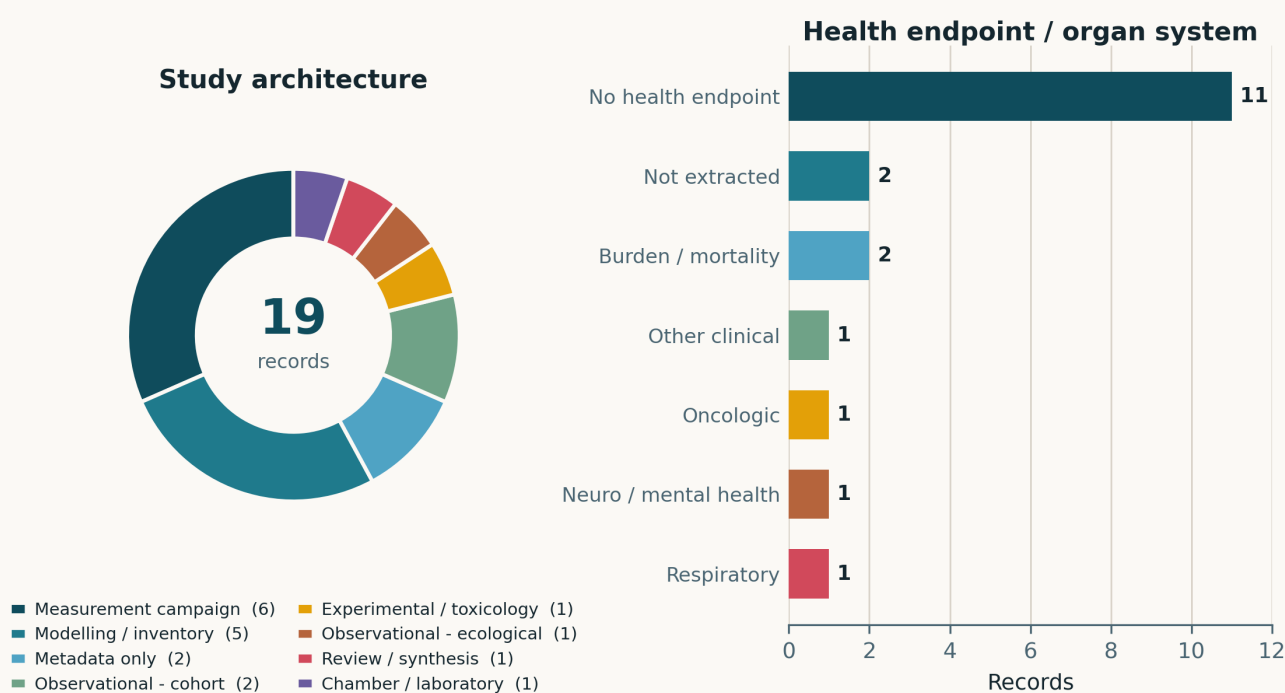


Fig. Left — study architecture. Measurement campaigns (6) and modelling/inventory work (5) together account for 11 of 19; primary human epidemiology contributes only 3 (two cohorts, one ecological analysis). Two records are *Metadata only*: they were surfaced through the Crossref journal sweep with no abstract deposited, so design, sample and endpoint could not be extracted and are not guessed. Right — health endpoint by organ system. Eleven of nineteen records carry no human health endpoint at all, which follows directly from the exposure/deposition weighting on the left.

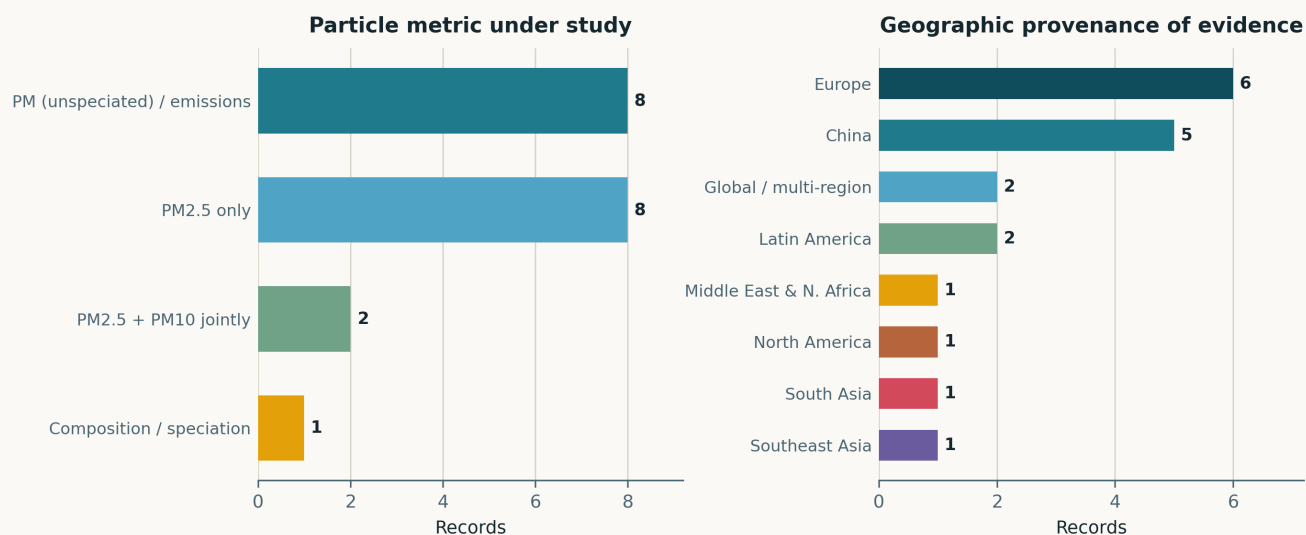


Fig. Left — particle metric. PM_{2.5} alone is the exposure in 8 records; another 8 are unspciated PM or emissions, which here mostly means deposited or wind-eroded material rather than a regulated size fraction. Only one record is composition-resolved in the register sense, though three more (Wei, Chen, Chai) are component-resolved in their analysis. Right — geographic provenance. Europe leads with 6 and China follows with 5; sub-Saharan Africa contributes nothing today, and the two Latin American records are both Mexican or Brazilian modelling rather than measurement.

Life-course windows addressed today (records may span several stages)

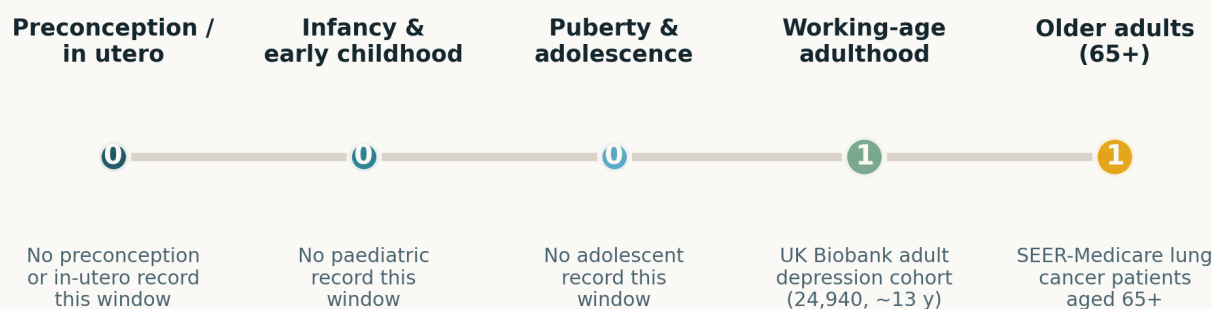


Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored at that stage. Two of the five windows are occupied and three are empty: the day produced no paediatric, adolescent or in-utero record at all. The two that are occupied sit at opposite ends — a working-age UK Biobank cohort and a ≥ 65 SEER-Medicare cohort.

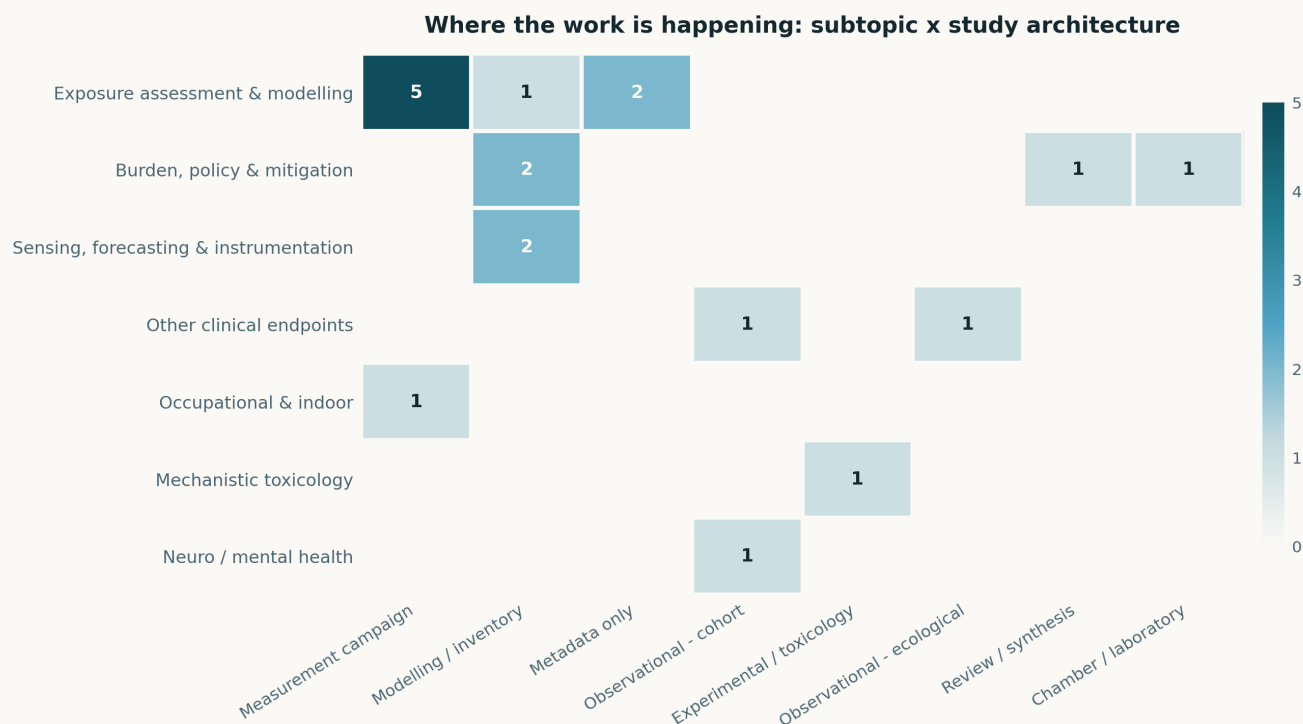


Fig. Subtopic × study-architecture cross-tabulation. The *Exposure assessment* row spreads across four architectures, which is what a broad but shallow day looks like. The two cohort cells (Neuro, Other clinical) are the only cells in the table that carry a human health endpoint together with a quantitative exposure contrast.

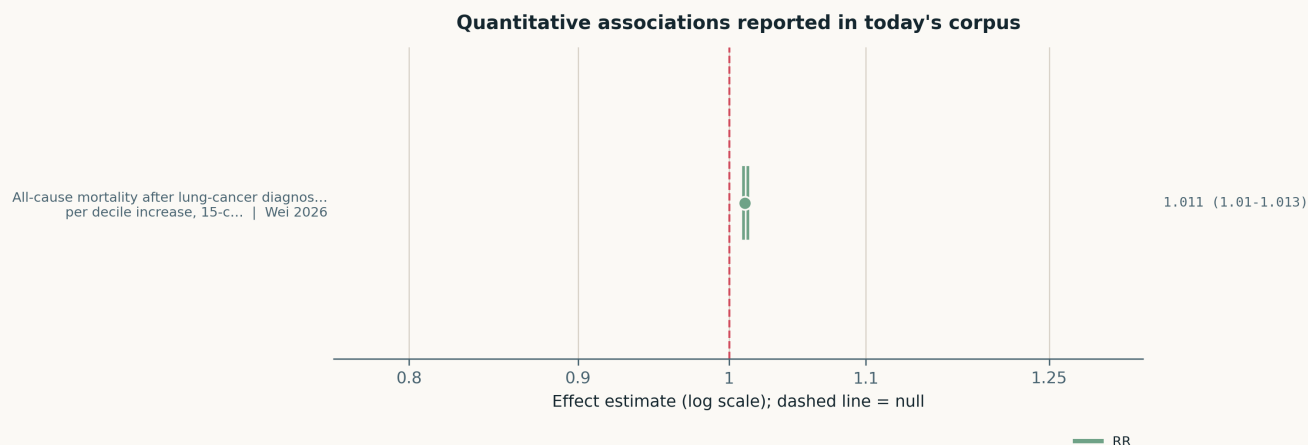


Fig. The single forest-plottable estimate in the corpus. The axis span is enforced at a minimum of $1.6\times$ in log space, and this issue is the reason. With one near-null estimate the axis previously auto-scaled to the data — a range of 1.5% — and drew RR 1.011 two-thirds of the way from the null line to the right edge. Nothing was false and everything read as a large effect. Chen et al.'s $PM_{2.5}$ -depression result is deliberately *not* plotted: the abstract reports "HRs up to 1.20" across six exposure variables with no interval attached, and placing an upper envelope on a shared risk axis would require inventing a confidence interval.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Exposure assessment & modelling

8 records

Chai et al. 2026 Environ Pollut

Role of ammonia, aerosol pH and liquid water content in secondary inorganic aerosol formation in suburban Nanjing

Hourly water-soluble ions and precursor gases, January–March 2021, coupled to ISORROPIA-II and WRF-Chem. SIA is **49.55% of PM_{2.5} mass** (mean 34.4 µg/m³), nitrate-dominated. Continuous ammonia excess fully neutralises the anion pool; rising aerosol liquid water enhances aqueous oxidation and drives nitrate and ammonium into the particle phase, while pH governs both semi-volatile partitioning and the sulfate pathway (transition-metal-ion catalysis dominant). Thermodynamically the system is **total-nitrate-limited**: PM_{2.5} responds minimally to moderate NH₃ cuts and needs >60% reduction to move, whereas cutting HNO₃ alone matches joint HNO₃+NH₃ control. The policy inversion — prioritise NO_x, treat NH₃ as a local adjunct — is the transferable result.

doi: 10.1016/j.envpol.2026.128998

Kan et al. 2026 Atmosphere

Short-term observations of airborne microplastics in Phnom Penh: concentrations, aerodynamic size distribution and polymer composition

First AMP observations for Cambodia, by µFTIR-ATR imaging, with meteorology, gaseous co-pollutants, water-soluble ionic tracers and HYSPLIT back-trajectories. Four independent 72-h periods: **0.97 ± 0.30 MP m⁻³ in TSP** (range 0.55–1.27) and **0.33 ± 0.10 MP m⁻³ in PM_{2.5}** (0.23–0.49); 134 particles identified in TSP, 46 of them inside the PM_{2.5} fraction. PE, PP and PET dominate, 52% fragments, 82% below 30 µm Feret diameter. The authors are explicit that *n* = 4 precludes source attribution — read the polymer split as a baseline, not an apportionment.

doi: 10.3390/atmos17080804

Joshi et al. 2026 (preprint) Research Square

Environmental drivers of foliar particulate accumulation, leaf structure and biochemistry in urban trees in Delhi

Ten native species, four contrasting Delhi sites, summer and winter 2025, with branch-level triplicates. Total leaf-deposited PM rose from **120.13 ± 30.99 to 423.69 ± 30.99 µg cm⁻²** between seasons, fine PM from 6.82 ± 1.32 to 25.75 ± 1.32. *Plumeria alba* retained most (414.92), *Azadirachta indica* least (140.82). Higher loads tracked ascorbic acid and foliar pH positively and carotenoids negatively, and were *not* explained by specific leaf area, leaf dry matter content or thickness; APTI showed no seasonal change. The useful negative result is that capture capacity and physiological tolerance are separate axes, so single-metric species screening is inadequate.

doi: 10.21203/rs.3.rs-10524655/v1

Alaimo et al. 2026 Atmosphere

Elemental composition and Pb isotopic signatures in pine needles from the industrialized Milazzo area

Forty *Pinus pinea* needle samples by ICP-OES/ICP-MS across 36 elements, 25 of them also for ²⁰⁶Pb/²⁰⁷Pb and ²⁰⁸Pb/²⁰⁶Pb. Enrichment factors above 10 for Cd, Cu, Mo, Sb, V and Zn against EF 0.5–2 for Al, Fe and Ti cleanly separate anthropogenic from geogenic contributions. Pb ratios of 1.153–1.192 and 2.063–2.108 indicate mixing of industrial emissions with local geological background and, notably, *limited* residual leaded-gasoline signal. A good template for isotope-assisted source apportionment where no monitoring network exists.

doi: 10.3390/atmos17080805

Lazo et al. 2026 J Environ Qual

Biomonitoring trace metal deposition with Ligustrum lucidum: Albanian fire incidents

Passive biomonitoring across 14 urban, industrial and rural sites in Elbasan, leaves collected two weeks after an industrial-waste fire and a landfill fire. Al, V, Fe, Cu, Cd, Pb and Zn showed low-to-moderate variability; Cr, Co, Ni, As and Sb were highly variable with industrial-site outliers, and cluster analysis separated the two groups. Method demonstration rather than exposure assessment — there is no control period and no airborne concentration — but it is the kind of post-incident evidence that is otherwise unobtainable.

doi: 10.1002/jeq2.70243

Yang et al. 2026 Atmos Chem Phys

Impacts of anthropogenic aerosols on a snowfall event in the Guanzhong Basin and surrounding areas

Cloud-resolving fully coupled WRF-Chem, with emission-perturbation experiments separating aerosol–radiation from aerosol–cloud interactions. The precipitation response reverses sign between the basin and its surroundings: inside the basin, excluding ARI gives a slight precipitation increase with emissions via enhanced ice-phase processes, and ARI itself increases precipitation once the emission increase passes a threshold (higher relative humidity, larger ice water path, better survival of falling ice); outside, precipitation falls with emissions as ACI suppresses the liquid phase. Spatial distribution is largely insensitive. Relevant to anyone using precipitation as a wet-deposition assumption in a polluted basin.

doi: 10.5194/acp-26-11923-2026

Gonçalves et al. 2026 Atmos Pollut Res

Extreme Saharan dust outbreak: air quality crisis in NW Spain (March 2022)

Metadata only. Surfaced through the Crossref journal sweep with no abstract deposited. Title, journal, DOI and authorship are verified; design, sample size and quantitative outcome are not extracted and are not inferred. Flagged for abstract retry on the next run.

doi: 10.1016/j.apr.2026.103183

Zheng et al. 2026 Atmos Res

Aerosol modulation of precipitation systems migrating from mountains to plains in Northern China during early autumn

Metadata only. Same provenance and same caveat as the entry above. The topic is adjacent to Yang et al. on this page, but nothing in the two can be compared until an abstract or full text is available.

doi: 10.1016/j.atmosres.2026.109289

Burden, policy & mitigation

4 records

Liao et al. 2026 Environ Res*Beyond mortality: multi-outcome assessment of health economic benefits from air pollution control in China*

Daily mortality, hospital admissions and outpatient visits across 18 Guangdong cities, 2015–2022, with generalized additive concentration–response models and valuation by value of statistical life plus direct medical cost. PM_{2.5} and NO₂ control delivered **10,305.35 and 1,017.00 million CNY** respectively; O₃ control returned *negative* net benefits on every outcome. The headline methodological number is that multi-outcome benefits of PM_{2.5} control **exceed mortality-only estimates by 31.79%** — i.e. the standard mortality-only accounting understates the case by roughly a third. Over 85% of mortality-related benefit accrues to the elderly.

doi: 10.1016/j.envres.2026.125501

Bosso et al. 2026 J Environ Manage*AIRTREE v2.0: predicting ecosystem services provided by Italian urban parks under climate change scenarios*

Multi-layer canopy model over 43 national-reforestation-programme sites under RCP4.5 and RCP8.5 to 2054, calibrated against five ICOS flux sites (Kling-Gupta efficiency up to 0.90). Under RCP8.5 net primary productivity rises 9% and evapotranspiration 8%, O₃ and NO₂ removal fall, and PM₁₀ **removal increases slightly**. Species responses diverge sharply — *Pinus halepensis* leads on PM removal, *Sorbus torminalis* on O₃ uptake, *Quercus pubescens* on carbon. Open-access tool; the species-resolved PM output is the part worth reusing.

doi: 10.1016/j.jenvman.2026.130738

Fares et al. 2026 Environ Pollut*Can urban green infrastructure influence the urban heat island effect and air quality nexus?*

Review arguing that the thermal and air-quality benefits of green infrastructure are almost never assessed together, and synthesising the interaction: shading and evapotranspiration lower leaf surface temperature, which promotes thermophoretic deposition of particles, alongside foliar uptake of gaseous pollutants and inhibition of secondary aerosol formation. Well-maintained greenspace outperforms engineered installations, whose effects are more localised. The limitations section is the honest part — BVOC emission, small effect sizes, species and climate-zone dependence, and canopy-induced night-time warming.

doi: 10.1016/j.envpol.2026.128960

Moghaddam et al. 2026 (preprint) Research Square*Reducing particulate matter emissions in landfills using novel dust suppressants*

Wind-tunnel factorial on soil from the Meyami landfill in Mashhad: three mesh sizes, four suppressants (compost, vermicompost, bentonite, nano-clay bentonite), three doses, two exposure times, two wind speeds. Mesh 20 was optimal across all suppressants (>85% reduction in wind-erosion rate); **compost at 10% dose reached 86.52%** and bentonite at 20% reached 80%, both at 4 m/s. A bench result on a single soil — no field validation and no airborne concentration measured — but the compost finding is operationally cheap enough to be worth a field trial.

doi: 10.21203/rs.3.rs-10431018/v1

Sensing, forecasting & instrumentation

2 records

Ng et al. 2026 Atmos Chem Phys*Impacts of droughts on biomass burning emissions, air quality and public health in the Amazon*

Satellite fire activity plus GFEDv4s emissions driving GEOS-Chem High Performance, 2010–2015, with paired fire-on/fire-off simulations. In non-drought years biomass burning contributes **34.0% (32.0–35.5) of dry-season CO and PM_{2.5}**; in drought years that rises to **55.4% (44.1–76.7)**. Fire-induced PM_{2.5} and ozone raise premature mortality (GEMM) by 6.0% and 18.6% in non-drought years, 8.9% and 24.4% in drought years — *at stable deforestation rates*, which is the point: the climate-driven amplification is separable from the land-use driver. For network design, the drought-year contribution range is wide enough that a fixed seasonal baseline will misestimate the fire fraction by a factor of two.

doi: 10.5194/acp-26-11835-2026

Legaria-Santiago et al. 2026 Atmosphere*Leveraging remote traffic data for local air pollutant estimation across London monitoring sites*

Four interpretable tree ensembles (Random Forest, Extra Trees, LightGBM, XGBoost) across six predictor scenarios, from remote traffic + meteorology + temporal variables alone up to including one and four neighbouring stations, benchmarked against ridge regression, spatial interpolation and a cross-site validation. Without neighbouring stations, NO₂ RMSE fell from **9.73–11.66 to 8.72–11.52 µg/m³** when traffic information was added, and SHAP shows traffic variables contributing at a level comparable to neighbouring-station measurements in traffic-dominated settings. The practical claim — remote traffic counts can partially substitute for network density — is demonstrated for NO₂; the PM channels are reported but not where the gain is.

doi: 10.3390/atmos17080806

Other clinical endpoints

2 records

Wei et al. 2026 Environ Res*Chemical composition of fine particulate matter and all-cause mortality of older lung cancer patients*

528,127 SEER-Medicare adults aged ≥65 diagnosed with lung cancer 2000–2019, followed annually; two-year moving-average exposure to 15 PM_{2.5} components from high-resolution spatiotemporal models linked by residential ZIP code; generalized weighted quantile sum regression with random holdouts, adjusted for demographics, histology, stage, first-course treatment, comorbidity and neighbourhood covariates. **RR 1.011 (95% CI 1.010–1.013) per decile increase across all components**. Contribution differences were modest but silicon, nitrate, vanadium, zinc and iron led — road dust, traffic, fossil-fuel and heavy-fuel-oil combustion. Exploratory subgroups suggest a stronger association in non-small-cell, later-stage and female patients.

doi: 10.1016/j.envres.2026.125538

Guerrero-Morales et al. 2026 Environ Toxicol Pharmacol*Spatial distribution of long-term ultrafine particle concentrations and systemic lupus erythematosus burden in central Mexico*

Municipal-level ecological analysis: SLE mortality 1998–2023 and hospitalisations 2008–2024 from national databases against satellite-model UFP for 2010–2019, using rate ratios, Jenks natural breaks and principal component analysis. National SLE mortality 0.23–0.86 per 100,000; municipal mean UFP roughly **1,500–24,000 particles/cm³**. Higher UFP and higher hospitalisation burden co-occur in metropolitan municipalities; mortality is heterogeneous and the authors say it should be read descriptively. Ecological design, no individual exposure, explicitly hypothesis-generating — but it is one of very few UFP–autoimmunity datasets from a middle-income country.

doi: 10.1016/j.etap.2026.105143

Neuro / mental health

1 record

Chen et al. 2026 Environ Toxicol Pharmacol*Integrated plasma proteomics and metabolomes reveal potential biomolecules linking PM_{2.5} to depression*

24,940 depression-free UK Biobank participants, exposure to PM_{2.5} and five constituents (ammonium, nitrate, sulfate, elemental carbon, organic carbon), Olink proteomics and NMR metabolomics, Cox models with mediation, functional enrichment and drug-repurposing prediction. **982 incident cases over a mean 13 years**; HRs reported “up to 1.20 per SD” with ammonium the strongest constituent. **140 proteins and 3 metabolites** were statistically significant mediators, mediation proportions up to 8.00% and 3.42%, enriched in lipid and unsaturated fatty acid metabolism, immune regulation and inflammation. Note that the abstract gives no interval for the 1.20 figure, which is why it does not appear on this issue’s forest plot.

doi: 10.1016/j.etap.2026.105144

Mechanistic toxicology

1 record

Yan et al. 2026 Front Pharmacol*PM_{2.5} induces lung injury via mtDNA-cGAS-STING-mediated macrophage M1 polarization*

21-day subacute PM_{2.5} mouse model plus bone-marrow-derived macrophages in vitro. Exposure produced pulmonary inflammation and tissue injury with macrophage accumulation and M1 polarisation; mechanistically, mitochondrial damage released mtDNA and activated cGAS–STING. The load-bearing experiment is the rescue: **pharmacological cGAS inhibition with RU.521 attenuated STING activation, M1 polarisation and lung injury in vivo**, which converts a correlative pathway into an interventional one. Subacute rather than chronic, and a single exposure route — the dosimetry is the usual weak point.

doi: 10.3389/fphar.2026.1902928

Occupational & indoor

1 record

Liu & Hu 2026 (preprint) Preprints.org*Multivariable determinants of indoor PM_{2.5} and infiltration dynamics in 48 residential buildings across eight cities of Northern China*

Paired indoor/outdoor PM_{2.5} at 5-minute resolution for seven consecutive days in each of four seasons across Beijing, Tianjin, Shijiazhuang, Taiyuan, Jinan, Zhengzhou, Xi'an and Anyang — **32,256 hourly observations from 48 residences**, with building characteristics, occupant behaviour and meteorology recorded alongside. A multivariable OLS model with cluster-robust standard errors ($R^2 = 0.879$) gives outdoor PM_{2.5} $\beta = 0.891$, window-open fraction $\beta = 0.572$, cooking $\beta = 0.049$ and **air-purifier operation** $\beta = -0.935$, all $p < 0.001$. Infiltration factor **0.28 ± 0.15**; outdoor sources contribute **57–69%** of indoor PM_{2.5}, peaking in the heating season; median deposition rate 0.53 h^{-1} against a median air change rate of 0.52 h^{-1} , exceeding 2 h^{-1} with windows open. Preprint, not peer reviewed — but the sample size and the seasonal design are both well beyond what this literature usually carries.

doi: 10.20944/preprints202608.1362.v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Chen et al.	Integrated plasma proteomics and metabolomes reveal potential biomolecules linking PM _{2.5} to depression	<i>Environ Toxicol Pharmacol</i>	Cohort	PM _{2.5} + 5 constituents (NH ₄ , NO ₃ , SO ₄ , EC, OC)	UK
Mechanistic toxicology					
Yan et al.	PM _{2.5} induces lung injury via mtDNA-cGAS-STING-mediated macrophage M1 polarization	<i>Front Pharmacol</i>	Animal + in vitro	PM _{2.5} , 21-day subacute exposure	Laboratory (mouse + BMDM)
Occupational & indoor					
Liu & Hu 2026	Multivariable determinants of indoor PM _{2.5} and infiltration dynamics in 48 residential buildings across eight cities of Northern China	<i>Preprints.org</i>	Measurement campaign	Paired indoor / outdoor PM _{2.5} ; infiltration factor $F_{inf} 0.28 \pm 0.15$	Northern China (8 cities)
Sensing, forecasting & instrumentation					
Legaria-Santiago et al.	Leveraging remote traffic data for local air pollutant estimation: a scenario-based machine learning study across London monitoring sites	<i>Atmosphere</i>	Machine-learning model	NO ₂ , PM ₁₀ , PM _{2.5} , O ₃ from remote traffic counts	London, UK
Ng et al.	Impacts of droughts on biomass burning emissions, air quality, and public health in the Amazon	<i>Atmos Chem Phys</i>	Satellite + chemical transport model	Biomass-burning PM _{2.5} and CO (GFEDv4s / GEOS-Chem HP)	Amazon, Brazil
Exposure assessment & modelling					
Alaimo et al.	Elemental composition and Pb isotopic signatures in pine needles (<i>Pinus pinea</i> L.) from the industrialized Milazzo area	<i>Atmosphere</i>	Deposition sampling	Atmospheric trace-element deposition; Pb isotope source apportionment	Milazzo, Italy
Chai et al.	Role of ammonia, aerosol pH and liquid water content in secondary inorganic aerosol formation in suburban Nanjing	<i>Environ Pollut</i>	Measurement campaign	PM _{2.5} secondary inorganic aerosol (mean $34.4 \mu\text{g}/\text{m}^3$)	Nanjing, China
Goncalves et al.	Extreme Saharan dust outbreak: air quality crisis in NW Spain	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Saharan mineral dust (not extractable)	NW Spain
Joshi et al. 2026	Environmental drivers of foliar particulate accumulation, leaf structural and biochemical responses in urban trees in Delhi	<i>Research Square</i>	Deposition sampling	Leaf-deposited coarse (10-100 μm) and fine (2.5-10 μm) PM	Delhi, India
Kan et al.	Short-term observations of airborne microplastics in Phnom Penh: concentrations, aerodynamic size distribution, and polymer composition	<i>Atmosphere</i>	Measurement campaign	Airborne microplastics in TSP and PM _{2.5} (uFTIR-ATR)	Phnom Penh, Cambodia
Lazo et al.	Biomonitoring trace metal deposition with <i>Ligustrum lucidum</i> : Albanian fire incidents	<i>J Environ Qual</i>	Deposition sampling	Fire-derived trace-metal deposition on leaves (ICP-MS)	Elbasan, Albania
Yang et al.	Impacts of anthropogenic aerosols on a snowfall event in the Guanzhong Basin and surrounding areas	<i>Atmos Chem Phys</i>	Modelling / inventory	Anthropogenic aerosol (ARI and ACI pathways)	Guanzhong Basin, China
Zheng et al.	Aerosol modulation of precipitation systems migrating from mountains to plains in Northern China during early autumn	<i>Atmos Res</i>	Metadata only (no abstract)	Aerosol-precipitation interaction (not extractable)	Northern China
Burden, policy & mitigation					
Bosso et al.	AIRTREE v2.0: predicting ecosystem services provided by Italian urban parks under climate change scenarios	<i>J Environ Manage</i>	Modelling / inventory	PM ₁₀ and PM _{2.5} removal by urban canopy	Italy (43 sites)

First author	Focus	Journal	Design	Metric	Region
Fares et al.	Can urban green infrastructure influence the urban heat island effect and air quality nexus?	<i>Environ Pollut</i>	Review / synthesis	PM deposition and thermophoresis onto green infrastructure	Global
Liao et al.	Beyond mortality: a multi-outcome assessment of health economic benefits from air pollution control in China	<i>Environ Res</i>	Burden / economic model	PM _{2.5} , NO ₂ and O ₃ concentration-response	Guangdong, China (18 cities)
Moghaddam et al. 2026	Reducing particulate matter emissions in landfills using novel dust suppressants	<i>Research Square</i>	Chamber / laboratory	Wind-eroded fugitive dust (wind-tunnel)	Mashhad, Iran
Other clinical endpoints					
Guerrero-Morales et al.	Spatial distribution of long-term ultrafine particle concentrations and systemic lupus erythematosus burden in central Mexico	<i>Environ Toxicol Pharmacol</i>	Ecological / spatial	Satellite-modelled UFP, 1,500-24,000 particles/cm ³	Central Mexico
Wei et al.	Chemical composition of fine particulate matter and all-cause mortality of older lung cancer patients	<i>Environ Res</i>	Cohort	15 PM _{2.5} chemical components, 2-year moving average	USA

Provenance and method

Entry window **2026-08-21** → **2026-08-21**, contiguous with the 20 August issue. Built at 22:1x local on 22 August under the 22:00 rule, as a one-day backfill. **Sources.** PubMed E-utilities on [EDAT] returned 10 unique records (6 shipped). Europe PMC on CREATION_DATE returned 28 unique records, of which **14 were archival PMC back-scans** — legacy scans with no title and a publication year decades outside the window, re-ingested with a creation date inside it — 7 were already in `seen.json`, and 6 shipped. The Crossref by-ISSN journal sweep returned 34 records and shipped 7. **OpenAlex returned HTTP 429 for the 19th consecutive run** and remains dead. arXiv returned nothing new under the 3-day recency screen. **Screening.** 46 records were screened out and written to `state/rejected.jsonl` with an individual reason each; the largest single reason is the Europe PMC back-scan class, followed by the Crossref sweep's non-airborne chemistry (PFAS, microplastics in water, sediment geochemistry) and four retraction notices. Two borderline exclusions are logged as borderline: cross-floor bioaerosol transmission in drainage stacks, and an occupant-comfort building-physics cluster. **Labels.** Subtopic, design, particle-metric and geography are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Two records carry no abstract at all and are marked *Metadata only* rather than being characterised by inference. Effect estimates are transcribed verbatim from abstracts. All DOIs were checked by `check_dois.py` before publication. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.